

Notes: Greenwood (JFE, 2005)

Short- and Long-Term Demand Curves for Stocks: Theory and Evidence on the Dynamics of Arbitrage

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1 Big picture

- **Question.** How do prices respond when many stocks simultaneously experience large, measurable, non-informational demand shocks? Do prices move only for the directly shocked stocks, or also for related stocks through arbitrageurs' hedging demand?
- **Core setting.** The paper studies the April 2000 redefinition of the Nikkei 225 index. Thirty stocks were added, thirty were deleted, and the remaining 195 index constituents were downweighted. Index trackers had to rebalance quickly, creating large mechanical purchases and sales.
- **Key empirical advantage.** The Nikkei index's unusual price-weighted construction makes the cross-sectional demand shocks measurable. The event is simultaneous, very large, and not plausibly driven by firm-specific information about fundamentals.
- **Theory.** Greenwood develops a multi-security limits-to-arbitrage model. Risk-averse arbitrageurs absorb a vector of demand shocks. Event returns are proportional to each security's contribution to the total risk of the arbitrage portfolio.
- **Main prediction 1.** For affected stocks, event returns are related to the covariance matrix of fundamental returns times the demand-shock vector, $\Sigma\Delta X$. A stock can move a lot not only because its own demand shock is large, but because it covaries with other stocks that are being bought or sold.
- **Main prediction 2.** Unaffected stocks should also move if they hedge arbitrageurs' positions in affected stocks. Their returns are related to $F'\Delta X$, where F captures covariance between affected and unaffected securities.
- **Main prediction 3.** Post-event returns should reverse the initial event returns. The stronger the initial price pressure, the larger the subsequent expected reversal.
- **Headline evidence.** Additions gained about 19% during the event week, deletions fell by about 32%, and the 195 remainders fell by about 13%, while TOPIX was nearly flat. At least half of the event-week returns reversed over the following 20 weeks.
- **Arbitrage result.** An arbitrage portfolio accommodating 1% of total rebalancing demand earned around yen 3 billion in less than 15 weeks. Scaled to the total event, this implies a large wealth transfer from constrained index trackers to arbitrageurs willing to bear short-run risk.
- **Economic takeaway.** Demand curves for stocks slope down at both short and intermediate horizons, not simply because individual stocks are illiquid, but because arbitrageurs face portfolio risk when absorbing correlated demand shocks.

2 Incremental contribution of the paper

- **Moves from one-stock price pressure to multi-stock price pressure.** Earlier index-inclusion studies often ask whether a single stock's demand curve slopes down. Greenwood generalizes the problem to a whole vector of simultaneous demand shocks, which is closer to many real institutional trades.
- **Introduces an arbitrage-risk sufficient statistic.** The paper shows that the relevant variable is not just own net purchases, but the stock's contribution to the risk of the arbitrage portfolio, $\Sigma\Delta X$. This connects demand-curve estimation to mean-variance portfolio theory.
- **Shows spillovers to unaffected securities.** A major conceptual contribution is that stocks with no direct demand shock can still have event returns because arbitrageurs use them as hedges. This extends demand-curve logic beyond the directly traded securities.
- **Provides a clean empirical event.** The Nikkei 225 redefinition produced a rare event in which hundreds of stocks experienced large, simultaneous, measurable, mechanically generated demand shocks. This gives the paper much stronger cross-sectional power than typical index-addition studies.

- **Links event returns and post-event reversal.** The model predicts that the same variable that explains event-week price pressure should explain long-run reversion. This ties short-run demand curves to medium-run arbitrage profitability.
- **Quantifies limits-to-arbitrage in a liquid market.** The event involves large public companies in a major equity market, yet prices move strongly and reverse slowly. The paper therefore shows that limits to arbitrage matter even outside thin, obscure, or highly illiquid assets.
- **Contributes to theories of comovement and sentiment.** By showing that a demand shock to some stocks affects related stocks through hedging, the paper provides a mechanism for excess comovement and demand-driven cross-sectional return patterns.
- **Institutional contribution.** The paper gives one of the clearest empirical demonstrations that index contracts and benchmark-tracking mandates can create large wealth transfers when trackers are forced to rebalance quickly.

3 Data and measurement

3.1 Event: April 2000 Nikkei 225 redefinition

- The Nikkei 225 index is price-weighted after adjusting each stock's price by its face value.
- On April 14, 2000, Nihon Keizai Shimbun announced a major redefinition of the index.
- Thirty high-technology and large-cap stocks were added, thirty smaller index constituents were deleted, and the remaining 195 constituents stayed in the index but received lower weights.
- The redefinition became effective at the market open on April 24, 2000.
- Because tracking-error-minimizing indexers had to rebalance between the announcement and effective date, the paper defines the event window as April 14 close to April 21 close.

Interpretation: The design separates demand pressure from fundamental news. Additions, deletions, and remainders are affected by a mechanical index rule rather than by a common shock to their cash flows.

3.2 Affected and unaffected stocks

- **Additions:** 30 stocks newly added to the Nikkei 225.
- **Deletions:** 30 stocks removed from the index.
- **Remainders:** 195 stocks that remained in the index but were downweighted.
- **Unaffected stocks:** 1,042 Japanese stocks that were not in the index before or after the redefinition and had sufficient price and volume data.
- The directly affected group represents a very large share of Tokyo Stock Exchange capitalization, making the event economically large enough to affect marketwide arbitrage portfolios.

Interpretation: The unaffected-stock sample is crucial. If unaffected stocks move in the predicted direction, the result cannot be explained by simple liquidity effects tied to own demand shocks.

3.3 Demand-shock measurement

- The paper estimates net purchases for each affected stock from the change in its Nikkei index weight times the total institutional holdings linked to the index.

- Total institutional holdings linked to Nikkei tracking are assumed to be about yen 2.4 trillion.
- Net purchases are expressed both in yen and as a fraction of market value.
- Additions generally have positive net purchases; deletions and most remainders have negative net purchases.

Interpretation: The unusual Nikkei weighting rule is what makes the demand vector observable. In most asset-pricing settings, demand shocks are unobserved or confounded with information.

3.4 Arbitrage-risk measurement

- The covariance matrix Σ is estimated from three years of daily pre-event returns.
- The key affected-stock regressor is $(\Sigma \Delta X)_i$, the contribution of stock i to the risk of the arbitrage portfolio.
- For unaffected stocks, the key regressor is $(F' \Delta X)_i$, where F is the covariance matrix between affected and unaffected stocks.
- Excess returns are raw stock returns minus beta times the TOPIX value-weighted market return.

Interpretation: The model's empirical novelty is to transform a raw demand vector into a portfolio-risk vector. A stock's price impact depends on how its demand shock changes arbitrageurs' total risk.

4 Models and empirical specifications

4.1 Arbitrageur demand

Arbitrageurs are myopic, mean-variance investors. Their demand vector is

$$N_t = \frac{1}{\gamma} [\text{Var}_t(P_{t+1})]^{-1} (E_t(P_{t+1}) - P_t), \quad (1)$$

where N_t is arbitrageur demand, P_t is the price vector, and γ is risk aversion.

Interpretation: Arbitrageurs absorb demand shocks only if expected returns compensate them for the risk of the positions. If arbitrageurs were risk-neutral or if fundamentals were certain, demand shocks would not move prices.

4.2 Market clearing after the demand shock

At the event date, index trader holdings change by the demand-shock vector u . Market clearing is

$$N_t = Q - u, \quad (2)$$

where Q is fixed supply. Substitution into arbitrageur demand gives

$$P_t = E_t(P_{t+1}) - \gamma \text{Var}_t(P_{t+1})(Q - u). \quad (3)$$

Interpretation: Prices adjust so that arbitrageurs willingly hold the new residual supply. The more risk arbitrageurs must bear, the larger the required price change.

4.3 Event returns for affected securities

The model implies that the demand-induced component of event returns is proportional to

$$\Sigma \Delta X, \quad (4)$$

where ΔX is the vector of net purchases expressed in yen and Σ is the covariance matrix of fundamental returns. The main cross-sectional regression is

$$r_{i,t^*} = \alpha + \beta(\Sigma \Delta X)_i + \eta_{i,t^*}, \quad \beta > 0. \quad (5)$$

Interpretation: Own demand matters, but only after accounting for covariance. A positive purchase can have smaller price impact if it hedges other positions in the arbitrage portfolio.

4.4 Event returns for unaffected securities

For securities not directly affected by the index change, the regression is

$$r_{i,t^*} = \alpha + \beta(F' \Delta X)_i + \eta_{i,t^*}, \quad \beta > 0. \quad (6)$$

Interpretation: This is the paper's strongest test against simple liquidity stories. Unaffected stocks have zero own demand shock, so their movement must come from covariance with the affected stocks.

4.5 Post-event reversal

For affected stocks, post-event excess returns over horizon k satisfy

$$R_{i,t^*,t^*+k} = \alpha_k + \lambda_k(\Sigma \Delta X)_i + \eta_{ik}, \quad \lambda_k < 0. \quad (7)$$

The model also implies a direct event-return reversal regression:

$$R_{i,t^*,t^*+k} = \alpha_k + \lambda_k r_{i,t^*} + \eta_{ik}, \quad \lambda_k < 0. \quad (8)$$

Interpretation: The initial event return is a risk premium paid to arbitrageurs. As the demand shock is absorbed and arbitrage capital earns compensation, prices partly reverse.

4.6 Liquidity-control specification

To separate arbitrage-risk effects from stock-specific liquidity effects, the paper estimates

$$r_{i,t^*} = \alpha + \beta \Delta X_i + c(\Sigma \Delta X)_i + d(\Delta X_i / Vol_i) + e(\Delta X_i / MV_i) + \eta_{i,t^*}. \quad (9)$$

Interpretation: If liquidity alone explained event returns, scaled own-demand variables should absorb the effect. The arbitrage-risk term remains strong, supporting the portfolio-risk interpretation.

5 Identification and empirical design

5.1 Why the Nikkei event is useful

- The redefinition is large, public, and mechanical.

- The demand shocks are simultaneous, which reduces concerns that macro time effects explain cross-sectional differences.
- The changes are not plausibly new information about future cash flows, especially for remainders whose weights fell mechanically.
- Cross-sectional variation in demand is generated by the index's construction rules, not by firms choosing to issue or repurchase shares.

Interpretation: The empirical design is closest to an event-study natural experiment in demand. It is not randomized, but the variation is unusually clean for a market-wide asset-pricing setting.

5.2 Main threats and responses

- **Information effects.** Additions may be viewed as positive news or deletions as negative news. The paper addresses this by emphasizing remainders and cross-sectional variation unrelated to future cash flows.
- **Liquidity effects.** Thinly traded stocks may have larger price impacts. The paper controls for demand scaled by trading volume and market value; the arbitrage-risk effect remains.
- **Market movements.** Excess returns subtract beta times TOPIX returns, using betas estimated from pre-event data.
- **Arbitrage entry.** The model assumes a fixed short-run arbitrageur capital stock. This is plausible because rebalancing occurred quickly, shorting constraints limited many investors, and the event happened around a holiday.

Interpretation: The paper's credibility comes from combining clean demand measurement, covariance-based predictions, unaffected-stock tests, reversal tests, and arbitrage-profit calculations.

5.3 What the design cannot fully prove

- The covariance matrix is estimated from historical returns and may not perfectly capture fundamental risk.
- The total amount of Nikkei-indexed institutional money is assumed rather than directly observed for every investor.
- Arbitrageurs' actual portfolios are not fully observed.
- The event is unique and unusually large, so external validity to small routine rebalancings is not automatic.

Interpretation: Even with these limitations, the event provides unusually strong evidence that demand shocks can move prices through multi-security arbitrage risk.

6 Section-by-section study guide

- **Introduction.** Explains why simultaneous multi-stock demand shocks matter and previews the three predictions: affected-stock price impact, unaffected-stock hedging spillovers, and reversal.
- **Section 2: Arbitrage with many stocks.** Derives how risk-averse arbitrageurs price a vector of demand shocks. The key object is $\Sigma\Delta X$.
- **Section 3: Nikkei 225 redefinition.** Describes the event, the affected groups, and construction of the net-purchase vector.

- **Section 4: Cross-section of event returns.** Tests whether event returns are explained by arbitrage-risk contribution and whether unaffected securities also move.
- **Section 5: Profitability of arbitrage strategies.** Shows that reversal generated large arbitrage profits, implying costly price pressure for index trackers.
- **Conclusion.** Interprets the event as evidence for downward-sloping demand curves, hedging spillovers, and limits-to-arbitrage.

7 Tables and figures

7.1 Figure 1: Chronology + Figure 2: Raw Event Returns

Read:

- Figure 1 defines the event timing: announcement on April 14, institutions begin rebalancing on April 17, bulk rebalancing on April 21, and effective redefinition on April 24.
- The event window is April 14 close to April 21 close; the post-event window starts April 24.
- Figure 2 Panel A shows that additions rise sharply during the event week while deletions and remainders fall sharply.
- Figure 2 Panel B shows that a large part of the event-week divergence later reverses.
- The TOPIX benchmark is nearly flat during the event week, so the cross-sectional effects are not just marketwide movement.

Detailed interpretation: The chronology is critical because index trackers had one week to trade before the new index became effective. The figure makes clear why the event return is not a one-day announcement effect: the pressure was generated by predictable institutional trading between announcement and implementation. Figure 2 then shows the pattern a demand-shock model should generate: additions, which indexers must buy, increase; deletions and remainders, which indexers must sell or downweight, decline. The later reversion indicates that the event-week prices were not simply permanent changes in fundamentals.

Economic message: The two figures jointly establish the event-study design. Prices move when forced demand arrives and later partially normalize.

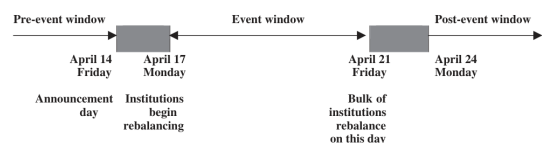


Fig. 1. Chronology. *Nihon Keizai Shinbun* announced the redefinition of the Nikkei 225 index around the close of the Tokyo Stock Exchange on Friday, April 14, 2000. The redefinition replaced 30 index securities and downweighted the remaining 195 securities. The change became effective when the market opened on April 24, 2000. To minimize tracking error, index traders would have submitted market close orders on Friday, April 21, 2000 or market open orders on Monday, April 24, 2000. Accordingly, event returns are defined on the window beginning on the close on April 14 and ending on the close on April 21. Weekly post-event windows are defined beginning on April 24.

(a) Figure 1: Chronology

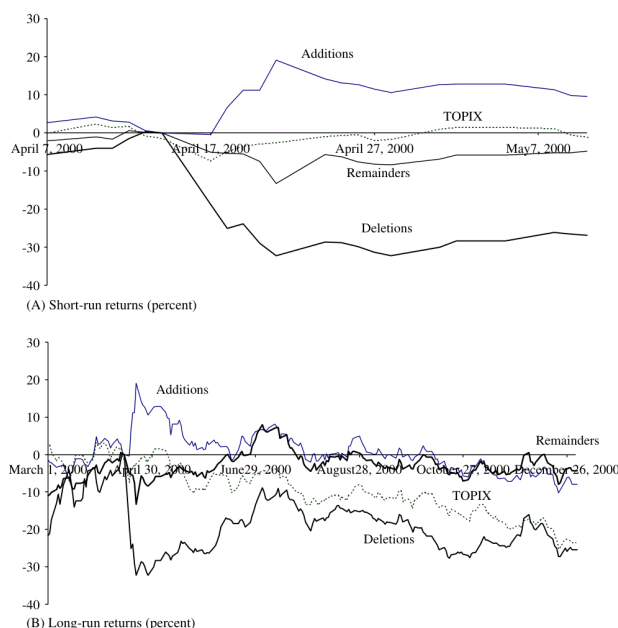


Fig. 2. Raw event returns. Buy-and-hold equal-weighted event returns for securities affected by Nikkei 225 inclusion. Additions are the 30 stocks added to the Nikkei 225 index, deletions are the 30 stocks deleted from the index, and remainders are the 195 stocks that remained in the index before and after the event. For each group of stocks, the figure plots equal-weighted buy-and-hold returns, set equal to zero on April 14, 2000. The figure also plots buy-and-hold returns of the Tokyo Stock Exchange value-weighted

(b) Figure 2: Raw event returns

7.2 Table 1: Summary Statistics, Event and Post-Event Returns + Table 2: Summary Statistics, Turnover

Read:

- Table 1 reports that during the event week additions gained 19.13%, deletions lost 32.29%, and remainders lost 13.35%.
- TOPIX lost only 1.18% during the event week, while unaffected stocks lost 2.08%.
- Post-event five-week and ten-week returns show strong reversal: additions fall while deletions and remainders rise.
- Table 2 shows turnover spiking during the event week, especially for deletions: weekly turnover for deletions was 13.48%, compared with a pre-event weekly mean of 1.48%.
- Additions also show large event turnover, confirming that price changes occurred during intense rebalancing activity.

Detailed interpretation: Table 1 quantifies the magnitude of the price pressure shown graphically in Figure 2. The numbers are too large to be dismissed as small indexing frictions. Table 2 links the returns to actual trading pressure: the return movements occur when turnover surges. The turnover evidence is important because the model is about demand shocks absorbed by arbitrageurs, not simply about announcement-date beliefs.

Economic message: The price changes and trading volume line up with forced rebalancing. The event creates an unusually clean measurement of demand-induced price pressure.

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Table 1
Summary statistics, event and post-event returns

Description	Date	Mean returns						
		Additions	Deletions	Remainders	All	Others	TOPIX	
Panel A: Equal-weighted daily returns (%)								
Thursday	Apr. 13	−1.99	1.78	−0.28	−0.24	−0.75	−2.48	
Announcement	Apr. 14	−0.50	−0.17	−0.12	−0.17	−0.82	−0.62	
	Monday	Apr. 17	−0.44	−18.81	−5.08	−6.15	−5.82	−6.12
	Tuesday	Apr. 18	7.26	−7.56	−0.26	−0.24	1.83	2.83
	Wednesday	Apr. 19	4.21	1.79	−0.08	0.64	1.43	1.47
	Thursday	Apr. 20	0.10	−6.77	−2.16	−2.43	1.76	0.57
Redefinition	Apr. 21	6.96	−4.85	−6.44	−4.67	−0.81	0.31	
	Monday	Apr. 24	−3.98	6.03	9.60	7.58	0.50	1.63
	Tuesday	Apr. 25	−0.87	−0.32	−0.67	−0.65	0.13	0.27
Panel B: Equal-weighted buy-and-hold weekly returns (%)								
Event week-1	Apr. 10-14	−2.11	6.13	2.48	2.37	−0.51	−1.43	
Event week	Apr. 17-21	19.13	−32.29	−13.35	−11.76	−2.08	−1.18	
Event week + 1	Apr. 24-28	−6.95	0.62	6.28	4.06	−0.35	0.90	
Event week + 2	May 1-5	2.06	6.20	2.89	3.18	3.48	3.26	
Post-event	Apr. 24-May 26	−13.98	9.63	9.47	6.73	2.29	−7.06	
five-week								
Post-event	Apr. 24-Jun. 30	−9.99	30.13	22.97	19.94	13.05	−2.60	
ten-week								
Post-event	Apr. 24-Sep. 8	−13.80	23.09	14.01	11.81	6.59	−7.60	
20-week								
Post-event	Apr. 24-Jan. 26	−23.16	12.24	7.49	4.44	6.39	−20.79	
40-week								
N		30	30	195	255	1,042	N/A	

Raw returns for securities affected by Nikkei 225 inclusion on April 24, 2000. Panel A shows average daily returns for each group of stocks starting one day before the announcement and ending two days after the close for which the rebalancing was effective. Panel B shows equal-weighted weekly returns starting two weeks prior to the event week and ending two weeks after. It also shows the buy-and-hold five-week, ten-week, 20-week, and 40-week post-event returns of a portfolio that is initially equal-weighted for each group of stocks. Post-event returns are measured relative to closing prices on April 21. Additions refer to the 30 stocks added to the index. Deletions refers to the 30 stocks deleted from the index. Remainders are the 195 stocks that remained in the index but whose weights changed. All is the 255 stocks in all groups. Others refers to a sample of 1,042 Japanese stocks not involved in the redefinition for which price and volume data were available during the event week. TOPIX is the Tokyo Stock Exchange value-weighted index, which includes all stocks traded on Section 1 of the exchange. All data are from Datastream. N/A = not applicable.

Table 2
Summary statistics, turnover

Description	Date	Additions	Deletions	Remainders	All	Others
<i>Panel A: Pre-event means (%)</i>						
Daily mean turnover		0.20	0.30	0.23	0.24	0.19
Weekly mean turnover		1.02	1.48	1.16	1.19	0.95
<i>Panel B: Daily event turnover (%)</i>						
Thursday	April 13	0.27	0.54	0.29	0.32	0.19
Announcement	April 14	0.27	0.82	0.35	0.40	0.18
Monday	April 17	0.83	1.68	0.34	0.55	0.21
Tuesday	April 18	0.58	2.03	0.29	0.53	0.20
Wednesday	April 19	0.65	2.12	0.29	0.55	0.16
Thursday	April 20	0.73	2.15	0.32	0.58	0.17
Redefinition at close	April 21	1.92	5.51	0.62	1.35	0.16
Monday	April 24	0.72	1.81	0.35	0.56	0.16
Tuesday	April 25	0.44	0.92	0.26	0.36	0.16
<i>Panel C: Weekly turnover (%)</i>						
Event week-1	April 10–14	1.14	2.38	1.34	1.44	0.95
Event week	April 17–21	4.70	13.48	1.86	3.56	0.91
Event week + 1	April 24–28	2.22	5.43	1.26	1.86	0.88
Event week + 2	May 1–5	0.45	1.06	0.38	0.47	0.48
Event week + 5 average	April 24–May 26	1.35	2.68	1.04	1.27	0.88
Event week + 10 average	April 24–June 30	1.27	2.59	1.23	1.39	0.97
Event week + 20 average	April 24–September 8	1.21	2.39	1.19	1.33	0.94
Event week + 40 average	April 24–January 26	1.20	1.84	1.10	1.20	0.84
N		30	30	195	255	1,042

Turnover for securities affected by Nikkei 225 inclusion on April 24, 2000. Turnover is defined as volume of shares traded divided by total shares outstanding. Panel A shows pre-event average equal-weighted daily and weekly turnover for the securities in each group. Panel B shows equal-weighted daily turnover starting one day before the announcement and ending two days after the close for which the rebalancing was effective. Panel C shows equal-weighted weekly turnover starting one week prior to the event week and ending two weeks after. Panel C also shows average weekly turnover for the five, ten, 20 and 40 week periods after the event. The exchange was closed Wednesday, May 3 through Friday, May 5 because of a national holiday. Turnover is correspondingly lower during this week. Additions refer to the 30 stocks added to the index. Deletions refers to the 30 stocks deleted from the index. Remainders are the 195 stocks that remained in the index but whose weights changed. All is the 255 stocks in all groups. Others refers to a sample of 1,042 Japanese stocks not involved in the redefinition for which price and volume data were available during the event week. All data are from Datastream.

Most stocks have a face value of 50, though some have face values of 5,000 or

(a) Table 1: Event and post-event returns

(b) Table 2: Turnover summary statistics

7.3 Table 3: Index Weight Changes Summary Statistics + Figure 3: Net Purchases of Stocks During Nikkei 225 Redefinition

Read:

- Table 3 explains why the Nikkei redefinition produced large net purchases and sales: the price-weighted index gives large weights to high-priced additions.
- Additions have no pre-event Nikkei weight and receive positive post-event index weight.
- Deletions have positive pre-event weight and no post-event weight.
- Remainders keep their index membership but experience lower average weights because the additions change the divisor and composition.
- Figure 3 Panel A shows the yen distribution of net purchases; additions are in the positive tail, while deletions and remainders are mostly negative.
- Figure 3 Panel B scales net purchases by market value, showing that some deletions experienced very large negative percentage demand shocks.

Detailed interpretation: Table 3 is the bridge from index mechanics to the demand vector ΔX . It explains why the event generated heterogeneous shocks, rather than a simple equal buy or sell for every stock. Figure 3 then shows the actual cross-sectional distribution of those shocks. This matters because the theory uses the whole vector of shocks, not just a group dummy for additions or deletions.

Economic message: The institutional structure of the Nikkei creates the experiment: demand shocks are mechanical, measurable, simultaneous, and heterogeneous.

Table 3
Index weight changes summary statistics

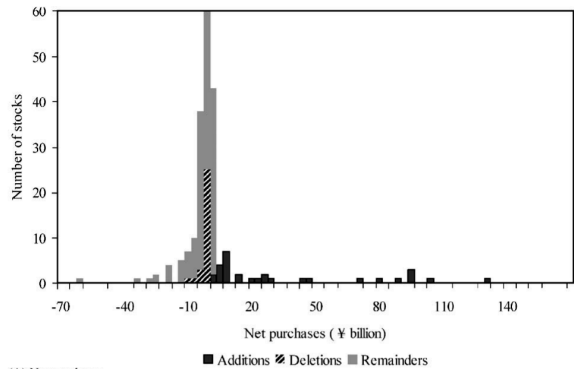
Sample	N	Price	Mean				
			Pre-event		Post-event		
			Nikkei 225 weight (%)	Nikkei Wgt Mkt Val Wgt	Nikkei 225 weight (%)	Nikkei Wgt Mkt Val Wgt	
			Face Value/50				
Additions							
Face value 50	26	5,096	5,096	N/A	N/A	1.46	3.15
Face value 5,000	1	1,111,000	11,100	N/A	N/A	3.18	3.88
Face value 50,000	3	1,691,667	1,691	N/A	N/A	0.48	0.25
All additions	30	210,583	4,956	N/A	N/A	1.42	2.88
Deletions							
Face value 50	30	261	261	0.13	9.14	N/A	N/A
Remainders							
Face value 50	189	960	960	0.46	2.57	0.28	2.09
Face value 500	4	4,288	429	0.21	0.42	0.12	0.34
Face value 5,000	1	1,550,000	15,500	7.45	3.85	4.44	3.14
Face value 50,000	1	1,490,000	1,490	0.72	0.07	0.43	0.06
All remainders	195	16,609	1,027	0.49	2.51	0.29	2.05

This table shows the correspondence between prices and Nikkei 225 weights of additions, deletions, and remainders and compares the index weights of these stocks to their market value as a percentage of the total market capitalization of the Nikkei 225. Stock price is the essential component in the Nikkei 225 calculation

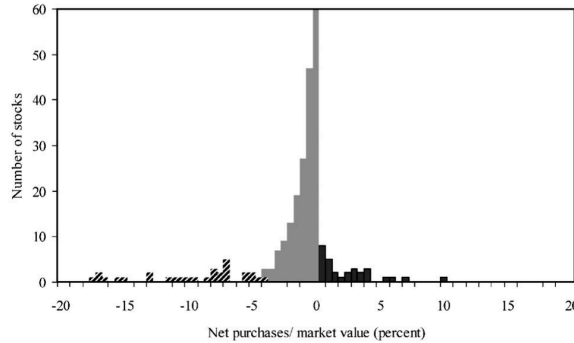
$$P_{Nikkei,t} = \frac{1}{D_t} \sum_{i=1}^{225} P_{i,t} F_i / 50$$

where D_t is the Nikkei 225 divisor, $P_{i,t}$ is the price of stock i on day t , and F_i is the face value of stock i ranging from 50 to 50,000. The table shows mean closing price on April 14, 2000, the day of the announcement of the Nikkei 225 redefinition. The table also reports mean stock price normalized by face value, which is the form in which prices enter the Nikkei 225 index calculation. The next two columns report the average weight in the Nikkei index and the average ratio of the Nikkei 225 weight to the weight the stock would take in the Nikkei if it were value-weighted. These two numbers are reported for both the pre-event index, which contains only the deletions and remainders, and the post-event index, which contains only the additions and remainders. All weights, including those of the additions, are based on prices on April 14, 2000. N/A = not applicable.

(a) Table 3: Index weight changes



(A) Net purchases



(B) Net purchases as a fraction of market value

Fig. 3. Net purchases of stocks during Nikkei 225 redefinition. Net purchases are defined as the change in

(b) Figure 3: Net purchases

7.4 Figure 4: Contributions to the Risk of Arbitrage Portfolio + Figure 5: Excess Event Returns and Arbitrage Risk

Read:

- Figure 4 plots the distribution of each stock's contribution to the risk of the arbitrage portfolio.
- Remainders account for much of the portfolio-risk variation because many of them are correlated with additions and deletions.
- Figure 5 Panel A shows a strong relationship between excess event returns and contribution to arbitrage risk among affected stocks.
- Figure 5 Panel B adds unaffected stocks and shows that they also line up with the risk-contribution prediction.
- The scatter is not simply an additions-versus-deletions story; within-group variation is also meaningful.

Detailed interpretation: Figure 4 converts raw net purchases into the model's central risk variable. A stock can have a positive or negative contribution to arbitrage risk even when its own direct demand shock is small, because it covaries with the rest of the demand vector. Figure 5 is the paper's most intuitive visual test: stocks that increase arbitrage portfolio risk require higher contemporaneous returns. Unaffected stocks moving in the same framework is especially important because it shows hedging spillovers.

Economic message: The demand curve is not stock-by-stock only. It is a portfolio object shaped by covariance and hedging.

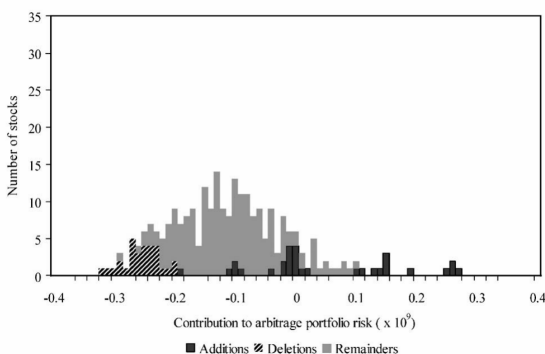
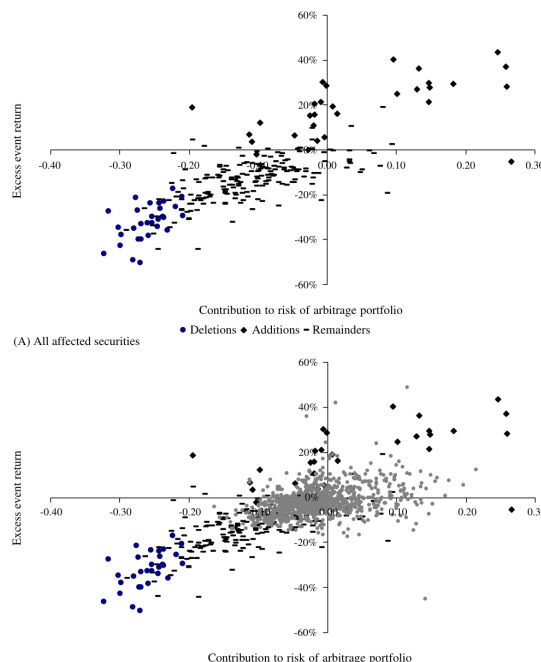


Fig. 4. Contributions to the risk of arbitrage portfolio. Histogram of contributions of each stock to the total risk of an arbitrage portfolio. For each stock, this is i th element of the product of the covariance matrix of fundamentals and the vector of yen denominated net purchases. The covariance matrix of fundamentals is estimated as the average covariance matrix of daily stock returns three years before the event. Net purchases are defined as the change in index weight times estimated total institutional holdings and are expressed in yen. They are computed using implied index weights on April 14, 2000 and assuming total institutional holdings of ¥2.4 trillion. Purchases are computed based on prices on April 14, 2000 and assuming total institutional holdings of ¥2.4 trillion. Additions, deletions, and remainders are marked

(a) Figure 4: Contributions to arbitrage risk



(b) Figure 5: Excess event returns and arbitrage risk

7.5 Table 4: Event Returns and Arbitrage Risk + Table 5: A Decomposition of Arbitrage Risk

Read:

- Table 4 shows that $(\Sigma\Delta X)_i$ explains a large fraction of event-week return variation.
- In the full affected sample, the univariate regression on arbitrage risk has an R^2 around 0.64, and adding raw net purchases raises it to about 0.68.
- The same logic works for remainders and deletions, and also for unaffected stocks through their covariance with affected stocks.
- Table 5 decomposes arbitrage risk into own variance-adjusted demand and hedging contribution.
- The hedging contribution is economically central, especially for explaining why results are not just own-demand effects.

Detailed interpretation: Table 4 is the core regression evidence. Raw net purchases matter, but the risk-adjusted portfolio contribution is the stronger theoretical variable. Table 5 goes one step deeper by separating a stock's own risk from its covariance with other shocks. The hedging term explains why some stocks with modest own shocks still move a lot and why unaffected stocks can move at all.

Economic message: Arbitrage-risk exposure, not just trading quantity, determines the slope of multi-stock demand curves.

Table 4
Event returns and arbitrage portfolio risk

N	a	$[t]$	b	$[t]$	c	$[t]$	R^2
<i>Panel A: All affected securities</i>							
255	-0.11	[-5.98]					0.00
255	-0.11	[-3.16]	0.004	[16.00]			0.27
255	0.01	[0.51]	0.002	[6.83]	1.06	[6.80]	0.68
255	0.03	[1.07]			1.20	[8.36]	0.64
<i>Panel B: Additions</i>							
30	0.20	[7.53]					0.00
30	0.13	[4.32]	0.002	[5.83]			0.44
30	0.12	[4.93]	0.003	[6.47]	-0.10	[-0.40]	0.44
30	0.18	[6.34]			0.56	[3.39]	0.28
<i>Panel C: Deletions</i>							
30	-0.31	[-5.59]					0.00
30	-0.34	[-5.92]	-0.011	[-2.51]			0.12
30	0.02	[0.25]	-0.008	[-1.70]	1.38	[3.60]	0.35
30	0.08	[0.98]			1.52	[4.15]	0.29
<i>Panel D: Remainders</i>							
195	-0.12	[-3.70]					0.00
195	-0.13	[-3.77]	-0.002	[-2.70]			0.04
195	-0.01	[-0.31]	0.002	[3.02]	0.92	[5.61]	0.45
195	-0.03	[-1.10]			0.82	[5.05]	0.43
<i>Panel E: Unaffected securities</i>							
1042	-0.01	[-0.57]					0.00
1042	-0.01	[-0.31]			0.43	[2.24]	0.13
<i>Panel F: All securities in sample</i>							
1,297	-0.03	[-1.37]					0.00
1,297	-0.03	[-1.37]	0.004	[16.00]			0.14
1,297	-0.01	[-0.26]	0.002	[9.27]	0.77	[4.85]	0.50
1,297	0.00	[-0.15]			0.84	[5.47]	0.45

Cross-sectional regressions of excess event returns on net purchases and the contribution of each stock to the risk of the arbitrage portfolio.

$$r_{it} = a + b\Delta X_i + c(\Sigma\Delta X)_i + \eta_{it}.$$

(a) Table 4: Event returns and arbitrage risk

Table 5
A decomposition of arbitrage risk

N	a	$[t]$	b	$[t]$	c	$[t]$	d	$[t]$	R^2
<i>Panel A: All affected securities</i>									
255	-0.11	[-3.16]	0.004	[16.00]					0.27
255	-0.11	[-3.20]	-0.001	[-0.66]	5.22	[4.59]			0.29
255	0.02	[0.74]	0.004	[6.63]	-1.40	[-2.08]	1.11	[6.95]	0.69
255	0.01	[0.47]			2.68	[12.72]	1.06	[6.63]	0.67
<i>Panel B: Additions</i>									
30	0.13	[4.32]	0.002	[5.83]					0.44
30	0.13	[4.19]	0.002	[2.28]	0.05	[0.06]			0.44
30	0.12	[4.81]	0.002	[2.89]	0.01	[0.01]	-0.10	[-0.40]	0.44
30	0.14	[5.81]			2.01	[7.57]	0.02	[0.10]	0.41
<i>Panel C: Deletions</i>									
30	-0.34	[-5.92]	-0.011	[-2.51]					0.12
30	-0.35	[-5.84]	-0.017	[-2.04]	2.97	[0.71]			0.13
30	0.03	[0.30]	-0.006	[-0.64]	0.39	[0.09]	1.40	[3.52]	0.35
30	0.06	[0.66]			-1.80	[-0.79]	1.48	[4.01]	0.35
<i>Panel D: Remainders</i>									
195	-0.13	[-3.77]	-0.002	[-2.70]					0.04
195	-0.13	[-3.90]	-0.005	[-1.48]	3.23	[0.80]			0.05
195	-0.01	[-0.34]	0.003	[1.34]	-1.55	[-0.40]	0.93	[5.71]	0.45
195	-0.01	[-0.42]			3.27	[3.43]	0.90	[5.42]	0.44

Cross-sectional regressions of excess event returns on net purchases, the own variance adjusted demand shock, and the hedging contribution to portfolio risk,

$$r_{it} = a + b\Delta X_i + c\sigma_i^2\Delta X_i + d\sum_{j \neq i}\sigma_{ij}\Delta X_j + \eta_{it}.$$

The independent variables are estimated net purchases of security i (ΔX_i), expressed in yen, idiosyncratic risk-adjusted net purchases ($\sigma_i^2\Delta X_i$), and the hedging contribution of security i to arbitrage portfolio risk ($\sum_{j \neq i}\sigma_{ij}\Delta X_j$). Panel A performs the regression using data on all securities affected by the index reallocation. This covers 30 additions, 30 deletions, and 195 remainders. Panel B performs the regression

(b) Table 5: Decomposition of arbitrage risk

7.6 Table 6: Event Returns and Arbitrage Risk Including Various Liquidity Controls

Read:

- Table 6 adds controls for net purchases scaled by trading volume and market value.
- The coefficient on arbitrage-risk contribution remains significant after these liquidity controls.
- The results hold across all affected stocks and within additions, deletions, and remainders.
- The full-sample specifications continue to show high explanatory power.

Detailed interpretation: This table addresses the main alternative interpretation: perhaps less liquid stocks simply move more when traded. The model's risk variable survives after controlling for demand relative to trading volume and market capitalization. This is important because a liquidity-only story cannot explain price movements for stocks with zero own demand shocks, nor can it explain the covariance structure emphasized by the model.

Economic message: Liquidity matters in markets, but here the key force is portfolio risk borne by arbitrageurs.

Table 6
Event returns and arbitrage risk including various liquidity controls

N	a	$[t]$	b	$[t]$	c	$[t]$	d	$[t]$	e	$[t]$	R^2
<i>Panel A: All affected securities</i>											
255	0.01	[0.51]	0.002	[6.83]	1.06	[6.80]					0.68
255	0.03	[1.08]			1.19	[8.31]	0.00	[1.83]			0.64
255	0.02	[0.73]			0.95	[6.71]			1.07	[4.62]	0.67
255	0.02	[0.74]			0.95	[6.70]	0.00	[1.13]	1.06	[4.57]	0.67
255	0.01	[0.20]	0.002	[6.40]	0.83	[5.50]	0.00	[0.89]	1.02	[4.30]	0.71
<i>Panel B: Additions</i>											
30	0.12	[4.93]	0.003	[6.47]	-0.10	[-0.40]					0.44
30	0.10	[2.75]			0.28	[1.58]	0.01	[5.46]			0.47
30	0.14	[4.65]			0.33	[1.80]			2.19	[4.11]	0.38
30	0.09	[2.65]			0.29	[1.60]	0.01	[3.88]	-0.45	[-0.53]	0.47
30	0.07	[2.09]	0.002	[4.25]	-0.15	[-0.62]	0.01	[2.93]	-0.40	[-0.47]	0.56
<i>Panel C: Deletions</i>											
30	0.02	[0.25]	-0.008	[-1.70]	1.38	[3.60]					0.35
30	0.03	[0.34]			0.93	[2.30]	0.00	[3.03]			0.41
30	0.09	[1.02]			1.42	[3.81]			0.31	[1.30]	0.32
30	0.03	[0.40]			0.91	[2.24]	0.00	[2.81]	0.16	[0.65]	0.41
30	-0.07	[-0.73]	-0.016	[-2.95]	0.51	[1.15]	0.00	[2.46]	0.75	[2.89]	0.55
<i>Panel D: Remainders</i>											
195	-0.01	[-0.31]	0.002	[3.02]	0.92	[5.61]					0.45
195	-0.03	[-1.10]			0.82	[5.05]	0.00	[-0.02]			0.43
195	-0.02	[-0.91]			0.71	[4.35]			1.56	[1.70]	0.44
195	-0.02	[-0.91]			0.71	[4.34]	0.00	[0.25]	1.57	[1.71]	0.44
195	-0.01	[-0.33]	0.002	[2.78]	0.82	[5.16]	0.00	[0.07]	1.17	[1.31]	0.45
<i>Panel E: All securities in sample</i>											
1,297	-0.01	[-0.26]	0.002	[9.27]	0.77	[4.85]					0.50
1,297	0.00	[-0.14]			0.84	[5.41]	0.00	[3.34]			0.46
1,297	0.00	[-0.21]			0.62	[3.77]			1.77	[5.31]	0.52
1,297	0.00	[-0.21]			0.61	[3.76]	0.00	[1.54]	1.76	[5.28]	0.52
1,297	-0.01	[-0.29]	0.002	[6.91]	0.59	[3.54]	0.00	[1.32]	1.55	[4.54]	0.55

Cross-sectional regressions of excess event returns on the contribution of each stock to the risk of the arbitrage portfolio and net purchases normalized by measures of liquidity

Table 6: Event returns and arbitrage risk including liquidity controls

7.7 Figure 6: Post-Event Excess Returns + Table 7: Post-Event Reversion

Read:

- Figure 6 plots ten-week post-event excess returns against event-week excess returns.
- Stocks that moved more during the event tend to reverse more afterward.
- Table 7 formalizes this with repeated cross-sectional regressions of post-event returns on event returns.
- The negative relation becomes stronger as the horizon extends, especially through ten to twenty weeks.
- Unaffected stocks also show reversion patterns consistent with the model.

Detailed interpretation: Figure 6 shows that event returns were not permanent information updates. The reversal is approximately cross-sectionally linear: the larger the initial price pressure, the more negative the subsequent return. Table 7 turns the visual pattern into estimates across horizons. The pattern supports the limits-to-arbitrage interpretation because arbitrageurs require compensation during the event and earn it as prices revert.

Economic message: Demand-induced returns are temporary risk premia, not permanent valuation changes.

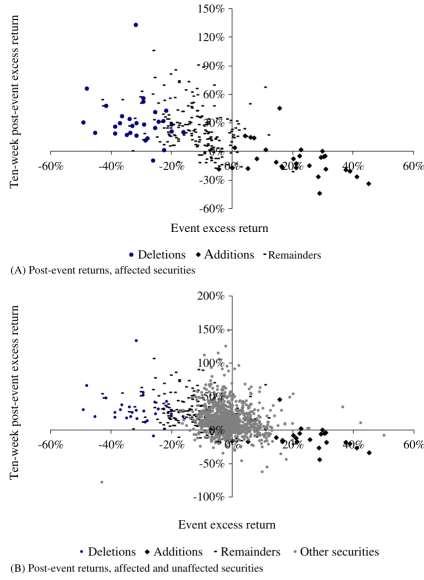


Fig. 6. Post-event excess returns. The figure plots buy-and-hold ten-week post-event excess returns for the 30 additions, 30 deletions, 195 remainders, and 1,042 unaffected securities against the excess return during the event week. Post-event excess returns are defined as the buy-and-hold return minus beta times the buy-and-hold return on the Tokyo Stock Exchange value-weighted index (TOPIX), where beta is computed using a market model. The horizontal axis is the excess event return. In Panel (A), the vertical axis is the ten-week post-event excess return and the sample includes only securities directly affected by the event. In Panel (B), the vertical axis is the ten-week post-event excess return and the sample is expanded to include 1,042 securities that did not experience weight changes during the event. Additions are marked with diamonds, deletions with filled circles, remainders with dashes, and all other securities with crosses.

Table 7
Post-event returns

Weeks after event	All affected securities (N = 255)				Additions (N = 30)				Deletions (N = 30)				Remainders (N = 195)				All unaffected securities (N = 1,042)				All securities (N = 1,297)			
(t)	α	β	λ	γ	α	β	λ	γ	α	β	λ	γ	α	β	λ	γ	α	β	λ	γ	α	β	λ	γ
1	-0.32	[-2.56]	-0.30	[-3.14]	-0.05	[-0.28]	-0.01	[-0.28]	-0.01	[-0.28]	-0.01	[-0.28]	-0.01	[-0.28]	-0.01	[-0.28]	0.04	[0.37]	-0.21	[-1.73]	-0.21	[-1.73]	-0.21	[-1.73]
2	-0.44	[-2.78]	-0.26	[-1.91]	-0.22	[-0.92]	-0.04	[-0.40]	-0.04	[-0.40]	-0.04	[-0.40]	-0.04	[-0.40]	-0.04	[-0.40]	-0.04	[-0.40]	-0.04	[-0.40]	-0.04	[-0.40]	-0.04	[-0.40]
3	-0.35	[-2.11]	-0.46	[-2.73]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]	-0.27	[-0.92]
4	-0.60	[-2.48]	-0.56	[-1.90]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]	-0.42	[-1.22]
5	-0.39	[-1.92]	-0.70	[-1.22]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]	-0.30	[-0.88]
10	-1.19	[-2.72]	-0.61	[-1.56]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]
15	-0.96	[-1.88]	-0.68	[-1.81]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]	-0.08	[-0.41]
20	-1.11	[-1.81]	-0.59	[-0.91]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]	-0.09	[-0.31]
40	-0.94	[-1.06]	-0.78	[-1.24]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]	-0.72	[-0.62]
1	0.01	[0.22]	-0.31	[-1.85]	-0.03	[-1.08]	-0.19	[-2.72]	-0.16	[-2.93]	-0.24	[-4.83]	0.00	[-0.14]	-0.53	[-5.58]	-0.00	[-0.22]	0.03	[0.37]	-0.00	[-0.13]	-0.22	[-2.58]
2	0.03	[0.75]	-0.26	[-1.17]	0.00	[-0.11]	-0.25	[-2.49]	-0.14	[-1.72]	-0.43	[-4.03]	0.02	[0.51]	-0.24	[-2.40]	0.01	[0.90]	-0.04	[-0.91]	-0.04	[-0.91]	-0.04	[-0.91]
3	0.02	[0.45]	-0.49	[-3.53]	0.01	[0.25]	-0.51	[-4.19]	-0.09	[-0.90]	-0.53	[-2.78]	0.01	[0.25]	-0.71	[-4.26]	0.04	[1.06]	-0.11	[-0.85]	0.03	[0.96]	-0.25	[-2.17]
4	0.02	[0.35]	-0.60	[-3.78]	-0.02	[-0.32]	-0.51	[-3.48]	-0.02	[-0.10]	-0.48	[-2.17]	0.02	[0.27]	-0.77	[-4.02]	0.04	[1.04]	-0.24	[-1.50]	0.04	[0.95]	-0.46	[-2.71]
5	0.00	[0.00]	-0.38	[-3.25]	0.00	[0.00]	-0.71	[-4.57]	0.02	[0.17]	-0.54	[-0.96]	0.01	[0.12]	-0.70	[-2.26]	0.02	[0.41]	-0.30	[-1.34]	0.02	[0.35]	-0.47	[-2.49]
10	0.12	[3.30]	-0.65	[-3.90]	0.02	[0.26]	-0.56	[-2.56]	0.14	[0.76]	-0.57	[-1.63]	0.12	[3.35]	-0.68	[-3.24]	0.13	[2.88]	-0.55	[-2.24]	0.13	[3.00]	-0.74	[-2.75]
15	0.04	[0.30]	-0.84	[-2.72]	0.02	[0.15]	-0.74	[-2.75]	0.00	[-0.02]	-0.85	[-1.08]	0.04	[0.35]	-0.96	[-2.50]	0.07	[0.93]	-0.56	[-1.97]	0.07	[0.94]	-0.71	[-2.10]
20	0.11	[0.86]	-0.79	[-2.23]	0.06	[0.47]	-0.62	[-1.98]	0.14	[0.57]	-0.57	[-1.14]	0.10	[0.82]	-0.87	[-2.17]	0.12	[1.37]	-0.69	[-1.42]	0.12	[1.30]	-0.88	[-1.80]
40	0.05	[0.20]	-0.79	[-1.98]	0.05	[0.26]	-0.64	[-2.13]	0.03	[0.10]	-0.62	[-0.80]	0.07	[0.37]	-0.81	[-1.73]	0.00	[0.03]	-0.66	[-1.38]	0.01	[0.04]	-0.83	[-1.53]

Estimates from repeated cross-sectional regressions of post-event excess returns on the excess event return and a constant. Excess returns are defined as raw returns minus beta times the returns on the Tokyo Stock Exchange value-weighted index (TOPIX) during the same period. Beta is estimated on three years of prior returns. Denoting the event week by t^* , event returns by r_{e,t^*} , and post-event excess returns between week t^* and $t^* + k$ as R_{e,t^*,t^*+k} , the table reports estimates from

$$R_{e,t^*,t^*+k} = \lambda_k r_{e,t^*} + \eta_k$$

and

$$R_{e,t^*,t^*+k} = \alpha_k + \lambda_k r_{e,t^*} + \eta_k$$

estimated for $k = 1, 2, 3, 4, 5, 10, 15, 20$, and 40 weeks after the event. In each regression, the independent variable is the excess event return. The bottom panel estimates also include a constant on the right-hand side. Standard errors allow for cross-sectional correlation.

(a) Figure 6: Post-event excess returns

(b) Table 7: Post-event reversion

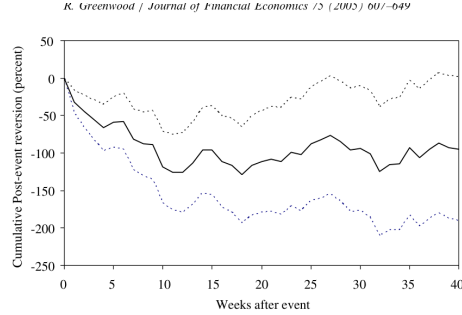
7.8 Figure 7: Tracing the Speed of Post-Event Reversion + Table 8: Post-Event Returns and Arbitrage Risk

Read:

- Figure 7 traces the fraction of the initial excess event return that reverses by each horizon.
- More than 20% of the event return reverses within one week, and substantial further reversal occurs over the next 20 weeks.
- Table 8 tests post-event returns against arbitrage-risk contribution rather than event returns.
- The signs are generally negative for affected stocks, consistent with the model's prediction that initial risk compensation reverses over time.
- The relation extends to unaffected stocks and all securities, implying that the Nikkei redefinition affected expected returns across much of the market.

Detailed interpretation: Figure 7 is helpful because it changes the perspective from cross-sectional sorting to time-series speed. The confidence intervals show uncertainty, but the overall pattern points to substantial reversion. Table 8 then links this reversion back to the model's structural risk measure, not simply the realized initial return. This confirms that the same portfolio-risk channel explains both the initial price movement and the subsequent expected return.

Economic message: Arbitrage capital earns compensation over weeks and months, not immediately. This slow reversal is a direct sign of limits to arbitrage.



Tracing the speed of post-event reversion. The figure plots estimates of the fraction of initial excess returns that are reversed by week k . These estimates come from repeated cross-sectional regression of post-event excess returns on the initial excess event return given in Table 7. The figure indicates the speed of reversion.

(a) Figure 7: Speed of post-event reversion

Table 8
Post-event returns and arbitrage risk

(k)	All affected securities (N = 255)					Additions (N = 30)					Deletions (N = 30)				
	a_k	[t]	λ_k	[t]	R^2	a_k	[t]	λ_k	[t]	R^2	a_k	[t]	λ_k	[t]	R^2
1	0.02	[0.60]	-0.21	[-1.44]	0.05	-0.07	[-2.57]	0.02	[0.14]	0.00	0.04	[0.51]	0.15	[0.41]	0.00
2	0.04	[1.08]	-0.25	[-1.26]	0.08	-0.06	[-1.44]	0.06	[0.27]	0.01	0.06	[0.48]	-0.01	[-0.01]	0.00
3	0.02	[0.38]	-0.51	[-2.05]	0.20	-0.07	[-1.51]	-0.36	[-1.25]	0.20	0.23	[1.52]	0.57	[0.90]	0.02
4	0.01	[0.11]	-0.70	[-2.44]	0.27	-0.11	[-1.92]	-0.37	[-1.11]	0.16	0.27	[1.56]	0.53	[0.72]	0.01
5	-0.01	[-0.23]	-0.71	[-2.21]	0.23	-0.12	[-1.90]	-0.50	[-1.34]	0.17	0.41	[2.12]	1.20	[1.47]	0.08
10	0.09	[1.03]	-1.10	[-2.44]	0.21	-0.07	[-0.79]	-0.45	[-0.87]	0.11	0.45	[1.66]	0.53	[0.46]	0.00
15	0.01	[0.13]	-1.06	[-1.92]	0.18	-0.10	[-0.96]	-0.66	[-1.03]	0.20	-0.03	[-0.10]	-1.14	[-0.80]	0.01
20	0.09	[0.78]	-0.93	[-1.45]	0.11	-0.05	[-0.37]	-0.35	[-0.48]	0.06	0.25	[0.64]	-0.27	[-0.17]	0.00
40	0.01	[0.05]	-1.14	[-1.26]	0.13	-0.10	[-0.54]	-1.02	[-0.98]	0.24	-0.12	[-0.23]	-1.35	[-0.58]	0.01
(k)	Remainders (N = 30)					Unaffected securities (N = 30)					All securities (N = 1,297)				
	a_k	[t]	λ_k	[t]	R^2	a_k	[t]	λ_k	[t]	R^2	a_k	[t]	λ_k	[t]	R^2
1	0.04	[1.34]	-0.23	[-1.41]	0.04	0.00	[-0.02]	0.34	[1.79]	0.08	0.00	[0.10]	-0.06	[-0.38]	0.00
2	0.06	[1.75]	-0.21	[-0.93]	0.03	0.03	[1.14]	0.41	[1.53]	0.10	0.03	[1.18]	-0.04	[-0.20]	0.00
3	0.04	[0.89]	-0.52	[-1.83]	0.14	0.04	[1.11]	0.01	[0.04]	0.00	0.04	[1.04]	-0.28	[-1.03]	0.05
4	0.03	[0.63]	-0.67	[-2.05]	0.17	0.04	[1.02]	-0.33	[-0.87]	0.02	0.04	[0.91]	-0.48	[-1.56]	0.09
5	0.01	[0.20]	-0.72	[-1.99]	0.14	0.02	[0.34]	-0.55	[-1.30]	0.05	0.01	[0.26]	-0.58	[-1.69]	0.11
10	0.12	[1.39]	-1.10	[-2.13]	0.12	0.13	[2.00]	-1.02	[-1.69]	0.07	0.12	[1.90]	-0.97	[-1.99]	0.12
15	0.05	[0.49]	-0.91	[-1.45]	0.08	0.06	[0.83]	-1.23	[-1.66]	0.10	0.06	[0.74]	-0.99	[-1.66]	0.12
20	0.13	[1.11]	-0.76	[-1.05]	0.04	0.12	[1.36]	-0.63	[-0.74]	0.02	0.12	[1.30]	-0.75	[-1.09]	0.05
40	0.04	[0.26]	-1.08	[-1.05]	0.07	-0.01	[-0.06]	-1.45	[-1.20]	0.07	-0.01	[-0.04]	-1.29	[-1.32]	0.11

Estimates from repeated cross-sectional regressions of post-event excess returns on the contribution of each stock to the risk of the arbitrage portfolio and a constant. Excess returns are equal to raw returns minus beta times the return on the Tokyo Stock Exchange value-weighted index (TOPIX) during the same period. Beta is estimated on three years of returns data prior to the event. Denoting the event week by t^* , risk-adjusted net purchases by $(\Sigma \Delta X)_t$, and post-event excess returns between week t^* and $t^* + k$ as R_{it^*, t^*+k} , the table reports estimates from

$$R_{it^*, t^*+k} = a_k + \lambda_k (\Sigma \Delta X)_t + \eta_{ik}$$

estimated for $k = 1, 2, 3, 4, 5, 10, 15, 20$, and 40 weeks after the event. Standard errors allow for cross-sectional correlation.

(b) Table 8: Post-event returns and arbitrage risk

7.9 Figure 8: Arbitrage Profits

Read:

- Figure 8 plots the value of arbitrage portfolios that absorb 1% of total rebalancing demand.
- A portfolio formed on April 21, after price pressure has occurred, earns around yen 3 billion within less than 15 weeks for only 1% of the total rebalancing demand.
- The dotted portfolio formed on announcement day first suffers a drawdown as pressure intensifies, then later recovers.
- Scaling to the full event implies a wealth transfer of roughly yen 300 billion from index trackers to arbitrageurs.

Detailed interpretation: The figure connects the paper's return regressions to an economically meaningful trading strategy. The gains are not abstract regression coefficients; they represent the price paid by benchmark trackers for immediate rebalancing. The initial drawdown also explains why arbitrage did not instantly eliminate the price pressure. Arbitrageurs had to bear substantial interim risk.

Economic message: The event reveals both the profitability and the risk of liquidity provision. Arbitrage was lucrative, but only for capital willing and able to absorb large temporary mispricing.

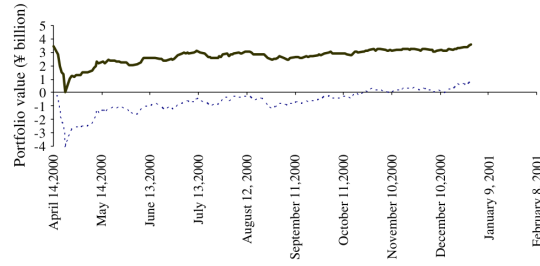


Fig. 8. Arbitrage profits. The figure shows the buy-and-hold value of the portfolio that is short 30 additions and long 30 deletions and 195 remainders in proportion to index weights based on closing prices on Friday, April 21, 2000. On this day, the portfolio is self-funded (has value zero) as the cost of the deletions and remainders exactly offset the proceeds from the additions. The portfolio corresponds to the holdings of an arbitrageur who absorbed 1% of the total demand induced by institutional rebalancing. The dotted line below shows the buy-and-hold value of the portfolio of an arbitrageur who absorbed 1% of total demand on April 14, the day the redefinition was announced. The demand shock in this case is calculated based on prices on that day. The vertical axis is ¥ billion (approximately ¥1 billion = US\$9 million). Institutional funds linked to the Nikkei are assumed to equal ¥2.4 trillion.

accommodates the demand shock on the date of the announcement (April 14), based

Figure 8: Arbitrage profits

8 Result map

- **Prediction 1: affected-stock event returns.** Event returns are increasing in $(\Sigma\Delta X)_i$. Table 4 and Figure 5 strongly support this.
- **Prediction 2: unaffected-stock spillovers.** Unaffected stocks move with $(F'\Delta X)_i$. Figure 5 Panel B and Table 4 Panel E support this hedging channel.
- **Prediction 3: decomposition into own and hedge components.** Table 5 shows that the hedging component is a central part of the story.
- **Prediction 4: liquidity robustness.** Table 6 shows that liquidity-scaled demand controls do not eliminate the arbitrage-risk effect.
- **Prediction 5: event-return reversal.** Figure 6 and Table 7 show a negative relation between event returns and post-event returns.
- **Prediction 6: arbitrage-risk reversal.** Table 8 shows that post-event returns are also related to the initial arbitrage-risk contribution.
- **Prediction 7: arbitrage profitability.** Figure 8 shows that the reversal generated large ex post profits for arbitrageurs.

9 Short summary

- Greenwood studies the April 2000 Nikkei 225 redefinition, a rare event with large, simultaneous, measurable demand shocks across many stocks.
- Thirty additions, thirty deletions, and 195 remainders are directly affected; 1,042 other stocks provide an unaffected benchmark and hedging-spillover test.
- The paper develops a multi-security limits-to-arbitrage model in which price impact depends on contribution to arbitrage portfolio risk.
- The key empirical variable for affected stocks is $\Sigma\Delta X$, the covariance matrix of returns times the net-purchase vector.
- The key empirical variable for unaffected stocks is $F'\Delta X$, their covariance with affected stocks times the affected-stock demand vector.

- Event-week returns are large: additions rise, deletions and remainders fall, while the broad market is nearly flat.
- Regression evidence shows that arbitrage-risk contribution explains a large share of cross-sectional event returns.
- Unaffected stocks also move in line with their hedging role, which is difficult to reconcile with liquidity-only explanations.
- Post-event returns reverse the event-week price pressure over the following weeks and months.
- An arbitrage portfolio that absorbs 1% of total rebalancing demand earns large profits, implying a large wealth transfer from constrained indexers to arbitrageurs.

10 Conclusion

10.1 Core conclusion

Greenwood shows that demand curves for stocks slope down in a multi-security setting and that the relevant price impact is determined by arbitrage portfolio risk. The April 2000 Nikkei 225 redefinition provides rare evidence because the event generates large, measurable, simultaneous demand shocks across hundreds of securities. Prices move strongly during the event and later reverse, consistent with limited arbitrage rather than permanent fundamental information.

10.2 Economic intuition

When index trackers must buy additions and sell deletions or remainders, arbitrageurs take the opposite side. But these arbitrageurs are risk averse and face uncertain fundamentals. They therefore require expected returns as compensation for holding a risky arbitrage portfolio. A stock's event return is high when absorbing its demand shock makes the arbitrage portfolio riskier. If the stock hedges other positions, its price impact can be smaller or even move in a direction not implied by own demand alone.

10.3 Incremental contribution and why it matters

The paper's incremental contribution is to replace the one-stock demand-curve view with a vector demand-curve framework. This is important because real institutional trades rarely affect a single security in isolation. Index rebalances, fund flows, swaps, portfolio transitions, and benchmark changes all generate multi-security demand vectors. Greenwood shows that the covariance matrix and hedging structure are essential for predicting price impact. This helps explain why demand shocks can create cross-sectional comovement and spillovers beyond the securities directly traded.

10.4 Implications for asset pricing

The results imply that prices can deviate from fundamentals even in large liquid markets when arbitrage capital is limited. This supports models of noise-trader risk, delegated arbitrage, and slow-moving capital. It also suggests that event returns in index-rebalancing studies should not be interpreted only as own-stock liquidity effects; they can reflect the full risk of the portfolio arbitrageurs must absorb.

10.5 Implications for index design and institutional investors

The Nikkei event shows that benchmark rules can impose large costs on investors who track mechanically. Immediate rebalancing transferred wealth to arbitrageurs. A more patient or staggered rebalancing process

could have reduced costs, although such policies may introduce tracking-error or agency concerns. The paper therefore links asset-pricing frictions to the design of index-tracking contracts.

10.6 Limitations

The event is unusually large and may not generalize perfectly to small routine rebalancings. The covariance matrix is estimated from historical returns. The paper does not directly observe every arbitrageur's position or every index tracker's constraint. The total amount of Nikkei-linked institutional capital is also estimated. These limitations do not overturn the main evidence, but they matter for applying the exact magnitudes elsewhere.

10.7 Takeaway

The paper's central lesson is that stock demand curves are portfolio demand curves. Demand shocks move prices through the risk borne by arbitrageurs, and this risk depends on covariance with all other securities being bought and sold. Arbitrage eventually pushes prices back, but only slowly and only after arbitrageurs are compensated for bearing risk.